

5.1.2 Out into space

Learning outcomes	Additional guidance
(a) <i>Describe and explain:</i> (i) changes of gravitational and kinetic energy (ii) motion in a uniform gravitational field (iii) the gravitational field and potential of a point mass (iv) angular velocity in rad s^{-1} (v) motion in a horizontal circle and in a circular gravitational orbit.	modelling the mass of a spherical object as a point mass at its centre
(b) <i>Make appropriate use of:</i> (i) the terms: force, kinetic and potential energy, gravitational field, gravitational potential, equipotential surface <i>by sketching and interpreting:</i> (ii) graphs showing gravitational potential as area under a graph of gravitational field versus distance, graphs showing changes in gravitational potential energy as area under a graph of gravitational force versus distance between two distance values (iii) graphs showing force as related to the tangent of a graph of gravitational potential energy versus distance, graphs showing field strength as related to the tangent of a graph of gravitational potential versus distance (iv) diagrams of gravitational fields and the corresponding equipotential surfaces.	
(c) <i>Make calculations and estimates involving:</i> (i) uniform gravitational field, gravitational potential energy change = mgh (ii) energy exchange, work done, $\Delta E = F\Delta s$; no work done when the force is perpendicular to the displacement, resulting in no work being done whilst moving along equipotentials (iii) $a = v^2/r$, $F = mv^2/r = mr\omega^2$ (iv) the radial components: $F_{\text{grav}} = -\frac{GmM}{r^2}$, $g = \frac{F_{\text{grav}}}{m} = -\frac{GM}{r^2}$ (v) gravitational potential energy $E_{\text{grav}} = -\frac{GMm}{r}$	Learners will also be expected to recall this equation M2.1